# Does the Early-Warning System Work? Prediction, Causation and Targeting in a Synthetic Student-Retention Cohort

Matthew C. Batitay

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Deliverable 3: Results and Discussion

**Everything in this deck is synthetic.**

Every number describes a generative model calibrated to cited aggregate statistics.
It is not a measurement of any real student, institution or programme.

# I use the data like an analyst, and the hidden truth only to grade my work

## The setup

- A synthetic cohort of **50,000 students** over 8 terms, calibrated to cited Philippine attrition: **67.1%** (high tier) and **45.9%** (typical tier).
- A **hidden** state (baseline vulnerability, acute shocks) drives the records *and* the dropouts. An analyst sees only eight exported tables.
- Support is assigned from prior-term signals, so it is **confounded**. The true effect is known: the dropout hazard is cut by **30%**.

## The rules I held myself to

- Analysis reads the **exported CSVs only**. The hidden truth is used in three labelled places, always to *grade* the result.
- **No leakage**: a feature for term $t$ uses entry attributes and terms 1 to $t-1$ only.
- Validation **grouped by student**; same folds for every model; nothing tuned on test data.
- Every added choice tagged `LOCKED`, `CALIBRATED` or `ASSUMED`.

Because I know the answer, I can say whether each method gets the *right* answer, not just whether it produces one.

# Six questions, one chain of evidence

**1. Is the generator valid?**

7 of 7 re-derived checks match the scorecard. The engine reproduces all 8 tables bit for bit.

**2. What does the data show?**

Attrition 67.1% vs 45.9% by tier, highest for low-income working students.

**3. Can I predict dropout?**

AUC **0.655** vs 0.567 for today's rule. Ceiling with perfect knowledge: 0.722.

**4. Does support work?**

The raw data say it **harms** (−58%). Done right: **+32%** against a true 30%.

**5. Who is missed?**

**64%** of dropouts occur with a clean record and no support.

**6. How robust is this?**

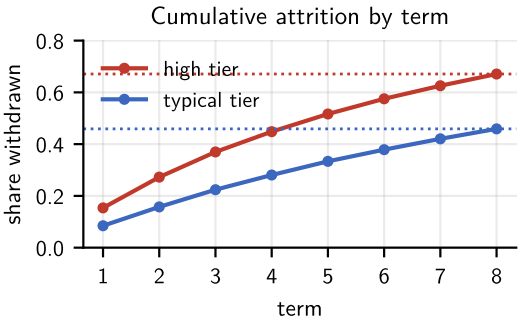
Holds across 11 recalibrated scenarios. IPW fails without overlap.

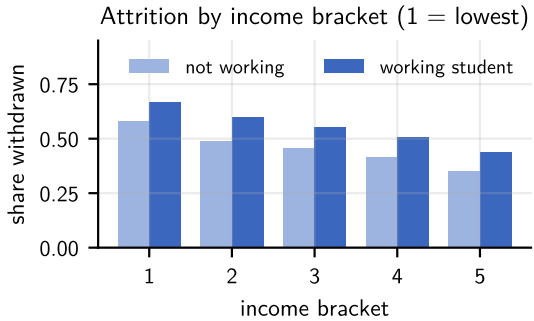
# The generator holds up when I re-derive its checks from the CSVs

| Check | Re-derived from the exported tables | Result |
|---|---|---|
| Tier attrition (calibrated) | High 67.1%, typical 45.9% (targets 67.1%, 45.9%) | pass |
| Correlation structure | Signals correlate (+0.39, −0.19, −0.18), but ≤ 0.009 once the hidden state is removed | pass |
| Confounding present | Naive estimate −57.7% (95% CI −64.6 to −50.6) excludes the true 30% | pass |
| Effect recoverable | Mantel–Haenszel 32.2% (24.3 to 39.3); Poisson 33.6% (25.1 to 41.2) | pass |
| Edge cases from the mechanism | 2,937 unflagged shock-driven dropouts; 774 falsely flagged terms | pass |
| Engine copy is faithful | All 8 tables regenerate with matching SHA-256; calibration loop returns the recorded values | pass |

**What this does not show.** It does not show the generator resembles any real institution. Calibration to the two tier targets is a convergence check, not independent validation. I did not re-run the generator's held-out Class-of-2023 check, which it labels *pass (fragile)*.

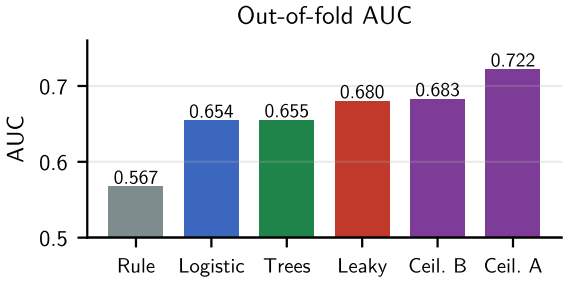
# What the data shows: two tiers, a steep gradient, and a selection effect

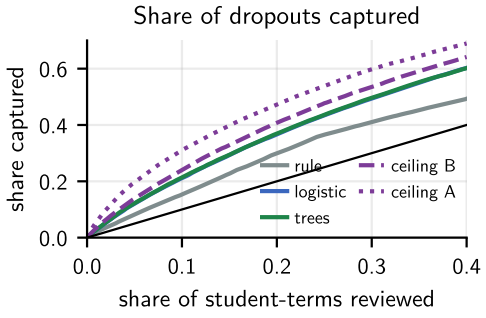



- Per-term dropout *falls* over time (high tier 15.4% to 12.1%): that is **selection**, since the most vulnerable leave first. It is not evidence that retention gets easier.
- Attrition runs from about **35%** (non-working, top income bracket) to **67%** (working, lowest). The gradient is built into the generator: a structure check, not a finding about the world.

# Early warning works, but only modestly




- **Boosted trees: AUC 0.655** (fold SD 0.005) against 0.567 for the signal-count rule. A logistic hazard gets 0.654, so the signal is nearly additive.
- Reviewing the top **10%** of student-terms captures **21.4%** of dropouts (rule: 15.3%); calibration is good in every decile.
- Drivers: the *running averages* of prior grades and attendance, not last term's flags.

# How good could any model be? The hidden truth sets the ceiling

| Model | AUC | PR-AUC | Top 5% | Top 10% | Top 20% |
|---|---|---|---|---|---|
| Signal-count rule | 0.567 | 0.104 | 8.2% | 15.3% | 29.9% |
| Logistic hazard | 0.654 | 0.139 | 12.1% | 21.1% | 36.7% |
| Boosted trees | 0.655 | 0.140 | 12.2% | 21.4% | 36.9% |
| *Leaky: + current-term signals (not deployable)* | 0.680 | 0.163 | 14.6% | 24.8% | 40.4% |
| *Ceiling B: knows baseline vulnerability* | 0.683 | 0.159 | 13.9% | 24.2% | 40.8% |
| *Ceiling A: true hazard* | 0.722 | 0.219 | 19.7% | 31.1% | 47.4% |

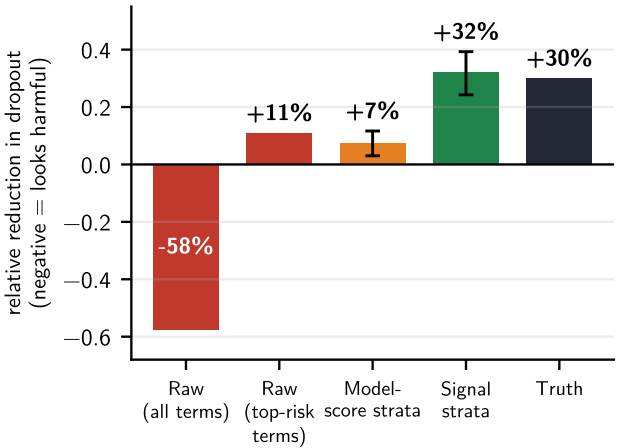
Out-of-fold, 247,739 student-terms (terms 2 to 8), event rate 8.0%. "Top $x$%" = share of dropouts captured by reviewing the top $x$% of student-terms.

**Much of dropout cannot be foreseen.** Even the true hazard gives AUC 0.722, because this term's shock is unknowable in advance.

**19.4% of dropouts** (4,762 of 24,573) happen in term 1, before any record exists. No term-based model can see them.

**Leaving out regional tier costs nothing** (AUC 0.655 vs 0.656), so excluding it on ethical grounds has no accuracy price here.

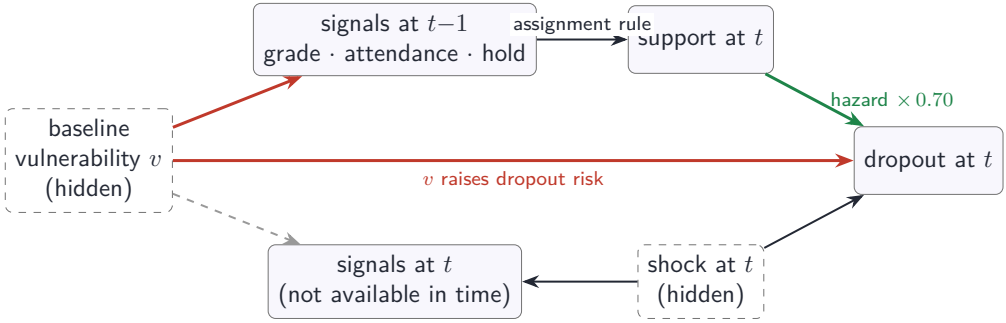
# A good predictor cannot tell me whether support works




- **Raw comparison: −58%.** Support goes to the students most at risk, so it looks harmful (confounding by indication).
- **Stratify on the model's score:** +7% (CI +3.0 to +11.7). Right sign, but it misses the truth.
- **Stratify on the assignment signals:** +32.2% (CI +24.3 to +39.3). Covers the true 30%.

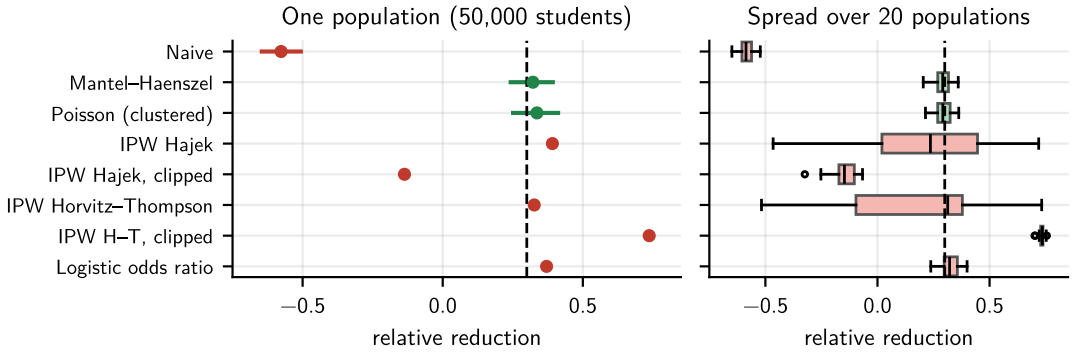
Ranking students and evaluating a programme need **different designs**.

# Why: a hidden common cause opens a backdoor path



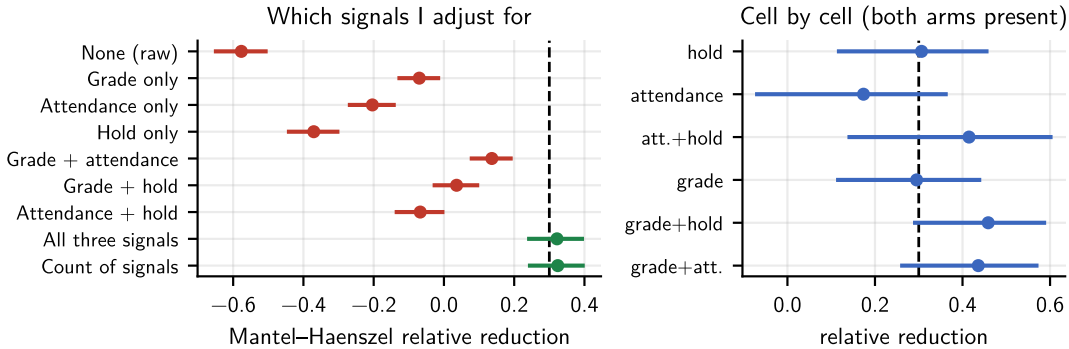

- The path support $\leftarrow$ signals$(t-1)$ $\leftarrow v \rightarrow$ dropout is open, which is why the raw comparison fails.
- Support depends *only* on the signals at $t-1$, so within a signal cell it is as good as random: **stratify on those signals**.
- The shock at $t$ is independent of everything before it, so it cannot confound.

# Estimator choice matters more than sample size



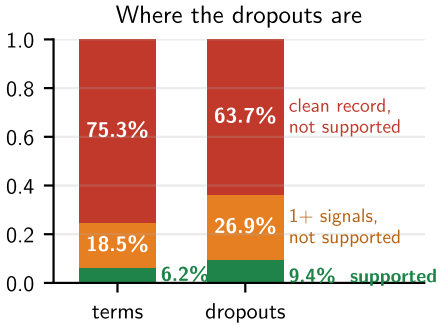

- **Mantel–Haenszel and Poisson** recover about 30% (SD about 4 points over 20 populations): one study resolves the effect to roughly ±8 points.
- **IPW swings from −47% to +72%** over the same populations: the supported arm's *effective* sample size is **43** of 15,293 (and I gave IPW the true propensity).

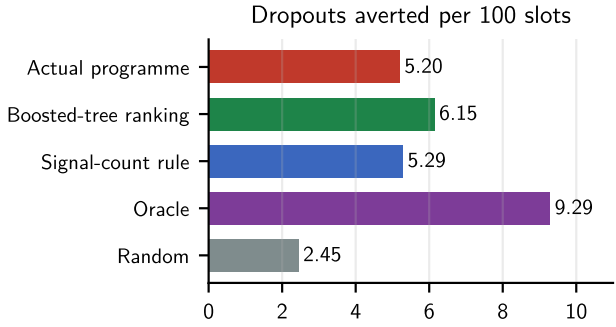
# Adjusting for only some of the referral signals fails




- Grade only −7.0%, attendance only −20.4%, hold only −37.1%. **Only the full set (+32.2%) or the count of signals (+32.4%) recovers the effect.**
- **Log the inputs to the referral decision, not just the flag**, or the programme looks harmful however well it works. Only six cells hold both arms, so effect heterogeneity is out of reach.

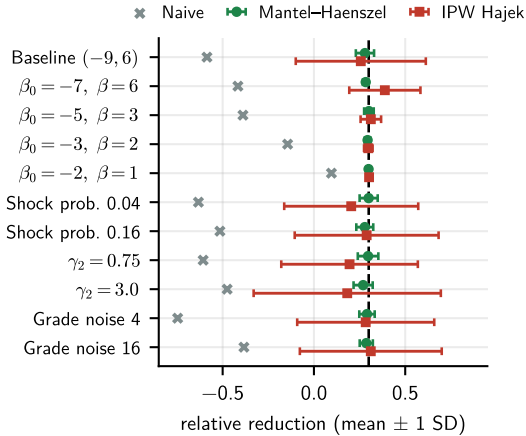
# Most dropouts happen where the programme is not looking




- **63.7%** of dropout events fall in clean-record, unsupported terms; only 9.4% fall in supported terms (6.2% of all terms).
- At equal capacity (15,293 slots) the model ranking averts an expected **940** dropouts against **796** for the actual rule (+18%), but only 66% of an unattainable oracle (1,421). This assumes a homogeneous 30% effect.

# The conclusions survive what I varied




- **11 scenarios**, each re-fitted to the same 67.1% / 45.9% targets, so differences are not just differences in attrition level.
- **Mantel–Haenszel stays near 30%** everywhere (replicate means 26.9% to 30.1%).
- **IPW instability is a positivity problem**: its SD falls from 35.6 points (default rule) to 1.1 (softest rule) as every signal cell gains both arms.
- Predictability is **capped by baseline vulnerability**: ceiling B stays within 0.659 to 0.686, while ceiling A ranges from 0.696 to 0.800 with shock strength.

# Recommendations: how to build and evaluate a retention programme

1. **Triage, do not judge.** Use the risk model to allocate attention, never to evaluate the programme.
2. **Evaluate with the assignment rule.** Log the referral inputs with the support flag; use Mantel–Haenszel or clustered Poisson. Expect ±8 points at 50,000 students.
3. **Build overlap into the rollout.** A small randomised share of referrals makes IPW and cell estimates usable.
4. **Pilot model-based allocation** head to head with the current rule before trusting the +18%.
5. **Look where the dropouts are,** not only where the flags are (64% have a clean record).
6. **Keep regional tier out** of the flagging model (no accuracy cost here).
7. **Handle IPW with care** under steep assignment: report effective sample size and largest weights.
8. **Do not act on these numbers.** The method is the point: validate against known truth, separate prediction from causation, and test how much each conclusion depends on assumptions.

## What these results are not

- **Everything is a property of the generator.** The prediction ceilings, the homogeneous 30% effect and the form of the confounding are assumptions, not measurements.
- **The causal problem here is easier than a real one.** Support never changes later grades (no carry-over), assignment has no unobserved drivers or caseworker discretion, the analyst knows the thresholds, and IPW gets the true propensity.
- **The targeting gain is conditional.** The policy value assumes the same 30% effect for everyone, so ranking by risk is optimal by construction.
- **The model comparison is narrow.** Two model families, fixed hyperparameters, one population; the sensitivity grid varies one parameter at a time with 10 replicates each.
- **Open items.** The generator's held-out check is fragile and was not re-run; the mechanism behind the biased score-stratified estimate is not investigated; term-1 dropouts (19.4%) fall outside both analyses.

Full list with status: notebook Section 12 (flags OF-1 to OF-12 from the generator, DF-1 to DF-9 new here).

# Take-aways

1. A risk model **ranks** students. It does not tell me whether the support **works**.
2. Evaluating support needs the **assignment inputs**, the right adjustment set, and **overlap** between those who were and were not supported.
3. Inside this synthetic world a better ranking buys **modestly** more, because much of dropout is unforeseeable and most of it happens where no flag is raised.

**Where to find it.** Notebook: `results-and-discussion/student-retention-analytics.ipynb` (executed copy: `.html`). It runs end to end from the exported CSVs, and every output carries a synthetic-data banner. Dataset and generator: `synthetic-data-generator/`. Specification: whitepaper v17.

# References


Gelman, A., Hill, J., and Vehtari, A. (2020). *Regression and Other Stories*. Cambridge University Press, Cambridge, U.K.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). *An Introduction to Statistical Learning with Applications in R*. Springer, New York, NY.

Law, A. M. (2015). *Simulation Modeling and Analysis*, 5th edition. McGraw-Hill, New York, NY.

Lopez-Muñoz, G. L., Gonzalez Serrano, C., and Sanchez-Ferreira, C. (2026). Student Dropout Prediction in Higher Education: A Systematic Review of Machine Learning Methods and Risk Factors. *Journal of Information Technology Education: Research*, 25, Article 20. https://doi.org/10.28945/5776

Peters, J., Janzing, D., and Schölkopf, B. (2017). *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge, MA.


The generator's calibration sources (CPBRD; EDCOM II) are listed in the whitepaper, Section 10.

# Backup A: policy values at equal capacity

| Policy (15,293 slots) | Averted | Per 100 slots | % of oracle | Slots per averted |
|---|---|---|---|---|
| Actual programme | 796 | 5.20 | 56.0% | 19.2 |
| Boosted-tree ranking | 940 | 6.15 | 66.2% | 16.3 |
| Signal-count rule, ties at random | 809 | 5.29 | 56.9% | 18.9 |
| Oracle (true untreated hazard) | 1,421 | 9.29 | 100.0% | 10.8 |
| Random | 375 | 2.45 | 26.4% | 40.8 |

Expected values inside the model: a slot is worth 0.30× the student-term's untreated hazard. Capacity is the number of supported student-terms in terms 2 to 8 (6.2% of 247,739). Total avertable if every term were supported: 6,097 dropouts.

- The oracle ranks by true untreated hazard, which includes *this term's shock*. That is why it is unattainable, and why a below-average-vulnerability student can still be its best pick.
- The signal-count rule with random ties performs like the actual rule, since the actual rule is that rule plus noise.

# Backup B: sensitivity grid in numbers

| Scenario | Supp. | Naive | MH | MH SD | IPW SD | AUC rule | AUC B/A |
|---|---|---|---|---|---|---|---|
| Baseline $(-9, 6)$ | 6.2% | −57.7% | +32.2% | 5.0 | 35.6 | 0.565 | 0.683/0.722 |
| $\beta_0 = -7, \beta = 6$ | 10.7% | −38.0% | +33.5% | 1.5 | 19.5 | 0.558 | 0.680/0.720 |
| $\beta_0 = -5, \beta = 3$ | 7.0% | −35.3% | +33.8% | 2.6 | 5.6 | 0.566 | 0.684/0.723 |
| $\beta_0 = -3, \beta = 2$ | 12.9% | −10.3% | +33.1% | 1.5 | 2.3 | 0.563 | 0.683/0.723 |
| $\beta_0 = -2, \beta = 1$ | 17.0% | +12.1% | +32.3% | 1.0 | 1.1 | 0.570 | 0.686/0.726 |
| Shock prob. 0.04 | 5.6% | −59.0% | +30.9% | 5.0 | 36.7 | 0.564 | 0.684/0.707 |
| Shock prob. 0.16 | 7.4% | −52.5% | +31.8% | 4.6 | 39.4 | 0.562 | 0.679/0.743 |
| $\gamma_2 = 0.75$ | 6.3% | −58.3% | +31.0% | 5.6 | 37.4 | 0.565 | 0.686/0.696 |
| $\gamma_2 = 3.0$ | 6.1% | −49.8% | +35.0% | 5.2 | 51.3 | 0.556 | 0.659/0.800 |
| Grade noise 4 | 4.9% | −74.9% | +32.2% | 4.2 | 37.5 | 0.561 | 0.684/0.722 |
| Grade noise 16 | 8.2% | −37.7% | +30.9% | 3.6 | 38.9 | 0.557 | 0.682/0.722 |

Each scenario recalibrated to 67.1% / 45.9%. Estimates are relative reductions (truth +30%). "MH SD" and "IPW SD" are SDs in percentage points over 10 replicate populations. "Supp." is the share of student-terms (terms 2 to 8) that receive support.